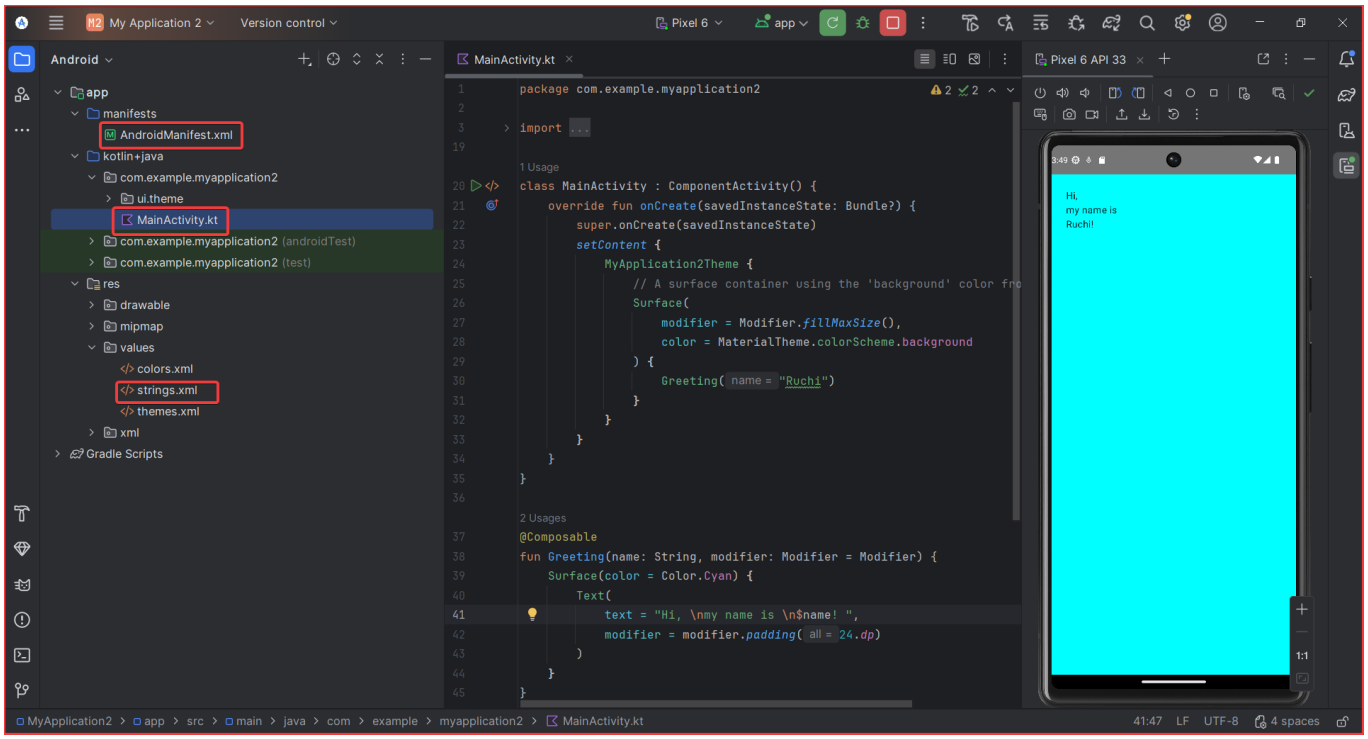
Task 1 — Project Setup, Run Log & Guided Walkthrough

1. Project Creation

Project Location:
java/com/example/myapplication2/MainActivity.kt



- Created a new project using **Empty Compose Activity** template.
- Default Kotlin and package settings were used.

2. Device / Emulator Setup

- Launched Android Emulator from Device Manager
- Device: Pixel 6
- API Level: 33 (Tiramisu)
- Android Version: 13.0

3. Short Checklist

- **Android Studio Version:** Ladybug | 2024.2.1
- **SDK Version Used:** API 33 (Tiramisu)
- **Emulator Name:** Pixel 6 API 33

4. Step-by-Step Log

Timestamp	Action	Result / Errors Encountered
10:15 PM	Launched Android Studio and selected "Empty Compose Activity"	Project initialized as My Application 2
10:20 PM	Initiated Gradle Sync	Success; Kotlin and Compose dependencies loaded
10:35 PM	Opened MainActivity.kt and viewed @Preview	Preview pane rendered "Hi my name is Ruchi!" correctly
10:44 PM	Clicked Run to launch on Pixel 6 Emulator	Error: Emulator slow to boot. Solution: Waited for cold boot; app deployed successfully

5. Technical Explanation

What does `setContent {}` do?

The `setContent {}` block is the entry point of the activity's UI.
It replaces the traditional `setContentView(R.layout.xml)` and defines the root Composable function that will be displayed on the screen.

What does `@Composable` mean?

The `@Composable` annotation tells the Kotlin compiler that the function is a UI-building block.
It allows the function to transform data into visual elements by describing the UI structure declaratively instead of using imperative UI code.

6. Lessons Learned

- **Gradle Dependencies:** First-run Gradle sync often downloads required SDK components like API 33, which may take time depending on network speed.

- **Preview vs Emulator:** The `@Preview` pane is useful for quick UI checks, but running the app on an emulator is necessary to verify actual behavior and performance.
- **Project Panel (Left Side)**
  - Point arrows to:
    * MainActivity.kt

    * AndroidManifest.xml

    * strings.xml
- **Emulator (Right Side)**
  - Label the device screen as **Running App on Pixel 6 Emulator**